

RESEARCH PROJECT PROPOSAL

Understanding Information Barriers to Regulatory Compliance Among Sri Lankan SMEs: A Data-Driven Investigation

Group: Enigmatrix | Faculty of Information Technology | University of Moratuwa
Level 04 Final Year Research Project | 2026

1. The One Core Problem

In One Sentence:

Sri Lankan SMEs want to comply — but they do not have access to the right information, at the right time, in the right language.

This is the single real-world problem driving the entire research project. Everything else flows from this one human truth:

Information Failure	What It Means
Right TIME	SMEs find out about regulatory changes weeks or months after they happen
Right INFORMATION	The guidance SMEs receive — from accountants, forums, social media — is often incorrect
Right LANGUAGE	Official documents are in English, but most SME owners operate in Sinhala or Tamil
Right AWARENESS	SMEs do not know they are heading toward non-compliance until a penalty arrives

2. Why This Is a Research Project — Not Just a System Build

A system build asks: "How do we build X?"

A research project asks: "We do not know if X is true, or how bad it is — let us find out."

System Build (What we had before)	Research Project (What we have now)
Built because it seemed useful	Built because research proves it is needed
Features decided by assumption	Features decided by measured findings
No validation of impact	Validated against real measured barriers
No novel contribution	4 novel datasets + 4 measurable findings
AI is the subject	AI is the tool — the real world is the subject

3. The Overarching Research Question

Main Research Question:

What are the information barriers that drive regulatory non-compliance among Sri Lankan SMEs — and how can they be systematically identified and addressed?

This question is:

- Genuinely unanswered — nobody has studied or measured this for Sri Lanka
- About the real world — not about how well AI works
- Requires all 4 modules to fully answer
- Generalizable — findings apply to any developing economy with similar regulatory fragmentation

4. The Four Research Modules

Each module investigates one specific information barrier. Together they form a complete picture of why Sri Lankan SMEs fail at compliance.

Module 1 — Regulatory Change Awareness Gap

Research Question:

Are regulatory changes reaching Sri Lankan SMEs in time to act — and what is the information lag between gazette publication and SME awareness?

The Real-World Problem

When a new regulation is published in the Government Gazette, how long does it take before an average SME owner knows about it? Days? Weeks? Months? Nobody has ever measured this for Sri Lanka. SMEs are getting penalized for missing deadlines they never even knew existed.

What You Investigate

- Measure the exact lag at every stage: Gazette → Official Portal → News → SME Awareness
- Identify which communication channels deliver regulatory information fastest
- Determine which types of regulatory changes have the longest awareness gap
- Find whether lag differs by SME sector, size, or location

Data Sources

Data Source	What You Collect	How You Get It
Government Gazette Archive	Publication dates of all regulatory changes 2018–2026	www.documents.gov.lk — all public PDFs
IRD / EPF / ETF / eROC websites	Secondary publication dates on official portals	Web scraping + manual archive
News Archives	First news coverage date of each regulatory change	Daily FT, LBO, Daily Mirror — searchable
SME Owner Survey	When SMEs actually became aware + which channel informed them	Google Forms via Chamber of Commerce, NEDA, LinkedIn

Procedures and Technologies

- Parse gazette PDFs using PyMuPDF / pdfplumber to extract publication dates and change content
- Train NLP classifier (TF-IDF baseline vs fine-tuned XLM-R) to categorize regulatory change types
- Calculate lag duration at each stage for every regulatory change
- Statistical analysis: mean, median, distribution of lags by change type and SME profile

- Channel effectiveness ranking — which source delivers information fastest

Novel Contribution

- First ever measured information lag dataset for Sri Lankan regulatory changes
- First ranked map of regulatory communication channel effectiveness in Sri Lanka
- First NLP classifier for Sri Lankan regulatory change categorization

The Solution Built

An automated gazette monitoring and SME alert system — validated by proving it delivers regulatory information significantly faster than current average awareness times.

Module 2 — Compliance Knowledge Accuracy Gap

Research Question:

How accurate is the compliance guidance Sri Lankan SMEs receive — and what is the gap between official regulatory requirements and what SMEs actually understand?

The Real-World Problem

When an SME files a VAT return, how correct is their understanding of what they need to do? When they ask their accountant or look at a Facebook group, how often is the answer wrong? Nobody has measured the accuracy of compliance guidance reaching Sri Lankan SMEs — or quantified how much misunderstanding exists.

What You Investigate

- Build a verified ground truth of all Sri Lankan compliance requirements across key regulatory domains
- Test SME owners and finance staff with specific compliance knowledge questions
- Calculate a compliance knowledge gap score — how far is SME understanding from official requirements?
- Measure accuracy of informal guidance (social media, forums, accountants) vs official sources

Data Sources

Data Source	What You Collect	How You Get It
IRD / EPF / ETF / eROC documents	All official compliance rules — exact rates, deadlines, formats	Public PDFs — verified by CA / tax professional
Compliance Knowledge Survey	SME responses to specific verifiable compliance questions	Google Forms — 80-120 SME owners and finance staff
Social Media and Forums	Compliance claims made in public groups	Facebook group scraping, Reddit Sri Lanka, forums
FactCheck.lk	Pre-labeled misinformation claims about regulations	Public website scraping

Procedures and Technologies

- Construct structured Q&A ground truth knowledge base from official documents — verified by expert
- Design and distribute a 40-50 question compliance knowledge test to SME owners
- Score responses against ground truth — calculate domain-specific accuracy rates
- Extract compliance claims from social media using spaCy NLP
- Verify each claim against ground truth — calculate informal guidance accuracy rate
- Build RAG system on verified knowledge base — evaluate using RAGAS faithfulness framework

Novel Contribution

- First measured compliance knowledge gap score for Sri Lankan SMEs
- First accuracy measurement of informal guidance channels in Sri Lankan regulatory context
- First Sri Lankan compliance Q&A benchmark dataset — verified against official sources

The Solution Built

A RAG-based multilingual compliance guidance system grounded strictly in verified official documents — validated by showing it produces more accurate answers than informal channels measured in the research.

Module 3 — Compliance Risk Invisibility

Research Question:

Which characteristics of Sri Lankan SMEs predict compliance failure — and can these signals be detected before a violation occurs?

The Real World Problem

SMEs do not know they are heading toward non-compliance until a penalty notice arrives. There is no early warning system. No one has studied which types of SMEs are most at risk or what behavioral patterns appear before a violation occurs. The risk is completely invisible until it is too late.

What You Investigate

- Which SME characteristics — sector, size, age, location — most strongly predict compliance failure?
- What behavioral patterns appear in the period before a compliance violation?
- Can risk be detected 1 month, 3 months, or 6 months before a violation occurs?
- Which ML approach generalizes best when labeled data is scarce?

Data Sources

Data Source	What You Collect	How You Get It
IRD Published Defaulter Lists	Records of actual tax violations with business type	IRD website — publicly published
Court Judgment Records	SME tax dispute cases with outcomes and timelines	www.lawnet.gov.lk — all public judgments
Central Bank Annual Reports	Sector-level SME financial health indicators over time	CBSL website — all public reports
SME Vulnerability Survey	Self-reported compliance failures with SME characteristics	Google Forms via Chamber of Commerce, NEDA
Synthetic Dataset	Realistic SME profiles calibrated to population statistics	SDV / CTGAN synthetic generation

Procedures and Technologies

- Combine real violation records, survey data, and synthetic data into unified labeled dataset
- Handle class imbalance using SMOTE oversampling
- Exploratory analysis: visualize compliance failure rates by sector, size, age, region
- Train and compare: Logistic Regression (baseline), Random Forest, XGBoost, LSTM
- SHAP analysis — identify which features most strongly predict non-compliance
- Temporal analysis — measure how early before violation risk signals become detectable

Novel Contribution

- First SME compliance risk prediction model for a developing economy using only public data
- First feature importance study identifying predictors of SME non-compliance in Sri Lanka
- Generalizable methodology for compliance risk prediction in any data-scarce regulatory environment

The Solution Built

An SME compliance risk early warning system with SHAP-explainable risk scores — validated by retrospective testing against known historical violations.

Module 4 — Regulatory Misinformation Spread

Research Question:

How significantly does tax regulation misinformation spread through Sri Lankan SME networks — and what makes certain misinformation more viral than accurate official guidance?

The Real World Problem

When an SME owner cannot find clear guidance from official sources, they turn to WhatsApp groups, Facebook, and online forums. Wrong information about tax regulations spreads rapidly through these channels. No one has measured how prevalent this misinformation is, how fast it spreads, or what makes it more believable than accurate information.

What You Investigate

- What percentage of tax regulation content on Sri Lankan social media is inaccurate?
- Which regulatory topics generate the most misinformation?
- Does misinformation spread faster and further than accurate content?
- What linguistic features make misinformation more viral than correct information?
- Does misinformation spike when new regulations are announced — correlating with Module 1 lag findings?

Data Sources

Data Source	What You Collect	How You Get It
Public Facebook Groups	Posts making compliance claims with engagement metrics	Facebook Graph API + manual collection from public groups
Twitter / X	Tax regulation discussions with retweet / reply data	Twitter Academic API with Sinhala / English keywords
Reddit Sri Lanka	Finance and tax discussion threads	Reddit API — public posts
FactCheck.lk	Pre-labeled Sri Lankan regulatory claims	Website scraping — publicly available
SME Survey (WhatsApp)	Anonymized forwarded messages about tax regulations	Collected via Module 1 / 2 survey instrument

Procedures and Technologies

- Collect and clean all social media posts containing tax regulation claims
- Translate Sinhala / Tamil posts using Google Translate API with manual verification
- Annotate dataset using Label Studio — accurate / partially accurate / misleading / false
- Calculate inter-annotator agreement using Cohen's Kappa
- Analyze spread patterns — compare engagement metrics of accurate vs inaccurate content
- Train misinformation classifier: compare fine-tuned XLM-R vs RAG-based claim verification vs GPT API
- Virality prediction model — identify linguistic features that predict high spread

Novel Contribution

- First annotated Sri Lankan tax regulation misinformation dataset
- First measurement of misinformation prevalence in Sri Lankan SME social networks
- First cross-lingual misinformation detection model for South Asian regulatory content

The Solution Built

A real-time claim verification interface — SME owners paste any compliance claim received from any source, and the system instantly checks it against verified official regulations with a cited verdict.

5. How All Four Modules Connect

The four modules are not separate tools — they are four investigations of the same problem feeding into one unified platform:

Module	Barrier Investigated	What It Feeds Into the Platform
1 — Awareness Gap	Regulatory changes not reaching SMEs in time	Gazette monitor → auto-updates knowledge base → triggers risk reassessment
2 — Knowledge Gap	Incorrect guidance being received by SMEs	Verified knowledge base used by all other modules
3 — Risk Gap	Compliance risk invisible until too late	Risk scores updated when Module 1 detects new regulations
4 — Misinformation Gap	Wrong information spreading faster than correct information	Claim verifier draws from Module 2 knowledge base

The Unified Platform Name:

SME Regulatory Intelligence Platform — An AI-powered system that delivers the right information, to the right SME, at the right time, in the right language.

6. Data Collection Summary

Module	Primary Novel Data	Supporting Public Data	Collection Method
1	SME awareness survey — when did they find out?	Gazette archives, news archives, IRD notices	Survey + web scraping + PDF parsing
2	Compliance knowledge test — what do SMEs actually know?	IRD / EPF / ETF / eROC official documents	Survey + document analysis + social media scraping
3	SME vulnerability survey — who fails and why?	Court records, IRD defaulter lists, Central Bank stats	Survey + public records + synthetic generation
4	Annotated social media misinformation dataset	FactCheck.lk, Facebook groups, Twitter, Reddit	Social media scraping + manual annotation

Important: No private enterprise data is required. All data is either publicly available, collectable through ethical surveys, or synthetically generated and calibrated against public population statistics.

7. Technology Stack

Layer	Technology	Used In
Data Collection	Scrapy, BeautifulSoup, Facebook API, Twitter API, Google Forms	All modules
PDF Parsing	PyMuPDF, pdfplumber	Module 1, 2
Data Processing	Python, Pandas, NumPy	All modules
NLP / Classification	HuggingFace Transformers, XLM-R, mBERT, spaCy	Module 1, 2, 4
RAG Pipeline	LangChain, ChromaDB, FAISS	Module 2, 4
ML Models	scikit-learn, XGBoost, TensorFlow / Keras	Module 1, 3
Model Interpretability	SHAP	Module 3
Annotation Tool	Label Studio	Module 4
Synthetic Data	SDV, Faker	Module 3
Translation	Google Translate API	Module 4
Evaluation	RAGAS, Cohen's Kappa, scikit-learn metrics	Module 2, 4
Visualization	Plotly, Seaborn, Matplotlib	All modules
Frontend	React.js, TailwindCSS	All modules
Backend	FastAPI	All modules
Database	PostgreSQL, ChromaDB	All modules

8. Novel Contributions — Why This Has Research Value

Contribution Type	What It Is	Why It Is Novel
New Dataset	Information lag dataset for Sri Lankan regulatory changes	Never measured or published before
New Dataset	Compliance knowledge gap scores for Sri Lankan SMEs	No benchmark exists for this
New Dataset	Sri Lankan tax misinformation annotated corpus	First of its kind for South Asian regulatory context
New Finding	Measured lag between gazette publication and SME awareness	Unknown — this research discovers it
New Finding	Accuracy rate of informal compliance guidance channels	Unknown — this research measures it

Contribution Type	What It Is	Why It Is Novel
New Finding	Top predictors of SME compliance failure in Sri Lanka	Unknown — this research identifies them
New Finding	Misinformation prevalence and spread patterns in SME networks	Unknown — this research quantifies it
New Framework	Public-signal-based compliance risk prediction methodology	Applicable to any data-scarce developing economy

9. The Complete Research Loop

Stage	What Happens
Problem	Sri Lankan SMEs are non-compliant due to 4 information barriers
Investigation	Each barrier is measured — lag duration, knowledge gap score, risk predictors, misinformation rate
Finding	Specific, quantified evidence of each barrier's severity — things nobody knew before
Solution	Each module directly addresses its measured barrier with a validated intervention
Validation	Solution is tested against the same measurements used to investigate the problem
Contribution	Proof that the solution reduces each barrier — with numbers, not assumptions

10. How to Present This to the Supervisor

What to Say:

Our research question is about a real human problem — why Sri Lankan SMEs fail at compliance despite intending to comply. We identify 4 specific information barriers as root causes and investigate each one empirically. Technology is our investigative instrument — not our research subject. Our findings will tell us something new about SME compliance behavior in Sri Lanka that nobody currently knows — backed by 4 novel datasets and 4 measurable research findings. No private enterprise data is required — all data is public, surveyable, or synthetically generated.

11. Individual Member Responsibilities

Member	Module	Individual Research Question	Novel Dataset
215075J — Mohamed M.R.I	Module 1 — Awareness Gap	What is the information lag between gazette publication and SME awareness?	Regulatory change lag timeline dataset
215007F — Ahamadh M.S.A	Module 2 — Knowledge Gap	How accurate is the compliance guidance Sri Lankan SMEs receive?	Compliance Q&A benchmark dataset
215008J — Ahamed T.I	Module 3 — Risk Gap	Which SME characteristics predict compliance failure before it occurs?	SME vulnerability and violation dataset
215019T — Cader Z.R	Module 4 — Misinformation Gap	How does tax misinformation spread through Sri Lankan SME networks?	Annotated misinformation corpus

12. Suggested Timeline

Phase	Period	Key Activities
Phase 1 — Problem & Literature	Feb 2026	Literature review, confirm research gap, finalize research questions per module
Phase 2 — Data Collection	Mar 2026	Gazette scraping, survey design and distribution, social media collection, ethics approval
Phase 3 — Analysis and Modelling	Apr 2026	Data analysis, model training, annotation, benchmark construction
Phase 4 — Platform Development	May 2026	Build unified prototype integrating all 4 module solutions
Phase 5 — Validation	Jun 2026	User testing, solution validation against research findings, evaluation metrics
Phase 6 — Write-up and Submission	Jul–Aug 2026	Research report, thesis chapters, demo preparation, viva rehearsal

— End of Research Proposal Summary —